👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.7

Reflect Points Across Axes

Lesson 7-9 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can reflect points across the x-axis and y-axis on the coordinate plane.

Language Objective: I can explain my work using the words reflection, x-axis, y-axis, and symmetry.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Reflection | **x-axis** | **y-axis** | **Symmetry** | **Integer** | **Perpendicular**

Level 3: try the blanks from memory first, then check the bank.

1 A flip of a point or shape over an axis to make a mirror image is a _____.

2 The horizontal number line on the coordinate plane is the _____.

3 The vertical number line on the coordinate plane is the _____ .

4 When two halves are mirror images of each other, the figure has _____ .

5 A whole number or its opposite, with no fraction part, is an _____ .

6 Two lines that meet at a 90° square corner are _____ .

Write About the Math

How do reflections help the game designer create a balanced, symmetric level?
¿Cómo ayudan los reflejos a la diseñadora del juego a crear un nivel equilibrado y simétrico?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Reflection** — *Reflexión*
- ☐ **x-axis** — *Eje x*
- ☐ **y-axis** — *Eje y*
- ☐ **Symmetry** — *Simetría*
- ☐ **Integer** — *Número entero*

Model: *Reflecting over the y-axis changes the ___ coordinate because ___.* — Student states reflecting over the y-axis flips the sign of the x-coordinate while the y-coordinate stays the same, so (3,2) becomes (-3,2).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The enemy reflected over the y-axis is at ___ because ___. The enemy reflected over the x-axis is at ___ because ___. Reflections help game designers because ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Reflection
2. **Notes 2:** x-axis
3. **Notes 3:** y-axis
4. **Notes 4:** Symmetry
5. **Notes 5:** Integer
6. **Notes 6:** Perpendicular

7. **Watch for this mistake:** *A common mistake in Reflect Points Across Axes is flipping the sign of the wrong coordinate — changing the x-coordinate when reflecting over the x-axis, instead of the y-coordinate (or vice versa for the y-axis). For example, to reflect (3, -5) over the x-axis, a student might flip the x-coordinate and write (-3, -5), but reflecting over the x-axis only changes the sign of the y-coordinate: (3, -5) becomes (3, 5). Before you submit, ask: 'Did I flip the sign of the coordinate that matches the axis I'm reflecting over — y for the x-axis, x for the y-axis?'*
8. **Try It 1:** C. (5, 3) — *Reflecting across the y-axis flips the sign of the x-coordinate only. (-5, 3) becomes (5, 3).*
9. **Try It 2:** B. The y-coordinate — *Across the x-axis the rule is (x, y) -> (x, -y), so only the y-coordinate flips sign. (8, -6) becomes (8, 6).*